

ASHOKA UNIVERSITY

Department of Computer Science

**FUEL UP+:
UNCERTAINTY-AWARE DEEP LEARNING FOR PERSONALIZED
GLUCOSE FORECASTING AND MEAL-SAFETY RANKING**

An Undergraduate Thesis

submitted in partial fulfillment of the requirements for the degree of
B.Sc. (Hons.) in Computer Science and Entrepreneurship with Research

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Certificate by the Supervisor

This is to certify that the thesis titled "Fuel Up+: Uncertainty-Aware Deep Learning for Personalized Glucose Forecasting and Meal-Safety Ranking" submitted by Arin Baswana, Student ID 1020221410, is a record of bona fide work carried out under our supervision at Ashoka University in partial fulfillment of the requirements of the undergraduate programme in Computer Science.

To the best of our knowledge, this thesis has not previously been submitted, in whole or in part, to any other university or institution for the award of any degree or diploma. It is recommended for evaluation following successful defence.

Prof. Lipika Dey

Prof. Partha Pratim Das

Date: _____

Date: _____

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Abstract

This thesis presents Fuel Up+, an extension of the Fuel Up nutrition-tracking capstone into an uncertainty-aware deep learning system for personalized glucose forecasting and meal-safety ranking. The core argument is straightforward: a point estimate of future glucose is not sufficient for meal-time decision support. If a model is used to compare candidate meals, it must communicate not only the most likely outcome but also the range of clinically plausible outcomes. To investigate this claim, three public continuous glucose monitoring datasets, HUPA-UCM, BrisT1D, and CGMacros, were harmonized into a common five-minute-grid pipeline using six-hour input windows and a sixty-minute prediction horizon. GRU and Transformer Encoder models were trained in absolute and delta target modes across progressively richer feature scenarios and were benchmarked against Persistence, Linear Drift, Ridge Regression, and LightGBM baselines. Two uncertainty mechanisms were then evaluated: Monte Carlo Dropout and Pinball-loss Quantile Regression. The empirical pattern is consistent across datasets. Of the two type 1 diabetes cohorts, the GRU in delta mode was the most reliable architecture, achieving 23.22 mg/dL MAE on HUPA S1 and 2.09 mmol/L on BrisT1D S3. On CGMacros, where meal annotation is much richer and glycemic dynamics are less insulin-dominated, the Transformer was competitive and marginally best in Scenario 2, while remaining near parity in Scenario 3. MC-Dropout produced narrow intervals but substantial under-coverage. Quantile Regression produced wider but materially better calibrated intervals, with observed 80% coverage of 64.4% on HUPA and 64.1% on BrisT1D. The final system exposes the forecast median together with its [q10, q90] band in a web dashboard that compares candidate meals through a 0-100 Safety Score. The contribution of the thesis is therefore twofold: a reproducible benchmarking pipeline across heterogeneous CGM datasets, and a practical argument that uncertainty-aware forecasting is more appropriate than scalar prediction for consumer-facing meal guidance.

1. Introduction

1.1 Context and Significance

Personalized nutrition is no longer only a conceptual ambition. It has become an engineering problem with accessible sensing infrastructure, mature sequence models, and a growing body of public metabolic data. The work of Zeevi et al. showed that post-prandial glucose response to the same meal can vary sharply across individuals, which undermines the assumption that population-level dietary advice is adequate for person-level decisions. At the same time, consumer nutrition applications still remain largely descriptive. They catalogue what a person has eaten, but they rarely estimate what a specific meal is likely to do next. This thesis takes that gap as its starting point (Zeevi et al.).

The motivation is both clinical and computational. Continuous glucose monitors now generate dense temporal records of metabolic state, while smartwatches and meal logs provide complementary behavioral context. Yet the relevant question for a user is not merely retrospective. It is prospective: given the last several hours of glucose, food intake, and activity, what is a reasonable forecast of the next hour, and how certain is that forecast? Fuel Up+ addresses that question by treating meal support as a forecasting problem rather than a logging problem.

1.2 From Fuel Up to Fuel Up+

The project began as Fuel Up, a student-built nutrition tracker designed to make meal logging easier in a university setting. The early application was functional but metabolically shallow. It could record calories and macros, but it could not model individual response. In practical terms, this meant that it could describe behavior without helping users choose differently. The turning point came when the project was reframed from descriptive nutrition tracking to predictive metabolic support.

Fuel Up+ is the result of that reframing. It ingests six hours of glucose, meal, and activity history, predicts glucose sixty minutes ahead, and attaches an uncertainty interval to that forecast. The prediction is then used in a meal-comparison interface that ranks candidate meals by expected glycemic safety. The aim is not to emulate a clinical decision-support system or an insulin controller. It is to build and evaluate a research prototype that translates forecasting output into a form a non-technical user can interpret before eating.

1.3 Thesis Contributions

This thesis makes four main contributions. First, it develops a harmonization pipeline that brings together three heterogeneous public CGM datasets into one evaluation framework with explicit protections against data leakage. Second, it compares GRU and Transformer architectures across multiple feature scenarios and target definitions, allowing architecture choice to be interpreted in relation to data richness rather than treated as a generic leaderboard question. Third, it evaluates two uncertainty mechanisms, MC-Dropout and Quantile Regression, on matched forecasting tasks and reports observed coverage rather than nominal coverage alone. Fourth, it connects the modeling work to a prototype user interface that turns forecasts into meal comparison through an interpretable Safety Score.

1.4 Clinical Stakes and Product Framing

The stakes of this problem are asymmetrical. A forecast error of 20 mg/dL is not equally consequential across the glucose range. A miss around 120 mg/dL may be tolerable; a similar miss near

hypoglycemia may change behavior in clinically dangerous ways. For this reason, the thesis treats calibration as more than a secondary reporting metric. If a model is intended to support meal choice, its uncertainty estimates must be sufficiently honest to indicate when the prediction should be trusted cautiously. That framing motivates the emphasis placed here on interval estimation, coverage, and clinically interpretable evaluation.

1.5 Roadmap of the Thesis

The thesis proceeds from motivation to method to interpretation. Section 2 outlines the physiological basis of glucose regulation and explains why uncertainty matters in this setting. Section 3 reviews the literature on CGM forecasting, uncertainty quantification, and personalized nutrition, and derives the gap analysis and research questions. Section 4 formalizes the problem statement and objectives. Section 5 describes scope, architecture, datasets, and evaluation design. Section 6 documents the implementation and its principal engineering challenges. Section 7 reports the empirical results and discusses what they imply. Sections 8 and 9 conclude the thesis and identify concrete next steps for extension and validation.

2. Background and Motivation

2.1 Clinical Physiology of Glucose Regulation

Blood glucose is regulated through a tightly coupled physiological system involving gastric emptying, intestinal absorption, pancreatic hormone release, hepatic glucose production, and peripheral glucose uptake. Any forecasting model trained on CGM data is therefore learning the observable trace of a multi-timescale control process. Carbohydrate ingestion initiates a rise in glucose, but the shape and timing of that rise depend on meal composition, recent activity, insulin state, circadian rhythm, and inter-individual metabolic variation. These dependencies explain why a sequence-modeling approach is more appropriate here than a static meal classifier (Ma et al.; Howey et al.).

In individuals without diabetes, endogenous insulin secretion and incretin effects usually constrain the post-prandial rise to a moderate excursion that resolves within a few hours. Fat, protein, and fiber alter the curve materially by slowing gastric emptying and changing the timing of glucose appearance in circulation. The result is that meals with identical carbohydrate totals may still produce different one-hour responses. This is important for Fuel Up+ because the sixty-minute prediction horizon is especially sensitive to timing effects rather than only to total absorbed carbohydrate (Ma et al.).

In type 1 diabetes, the problem becomes harder. Insulin is delivered exogenously, and the interaction between carbohydrate absorption and subcutaneous insulin uptake is noisy even when dose and meal are known. Rapid-acting analogues have broadly predictable onset and duration, but their exact kinetics vary with injection site, local blood flow, and recent exercise. A forecasting model working on T1D data must therefore approximate a process in which an important share of the variance remains unobserved. This is one reason uncertainty estimation is central rather than optional in the present thesis (Howey et al.).

A distinct circadian pattern, the dawn phenomenon, drives glucose upward between roughly 4:00 and 8:00 AM due to cortisol and growth-hormone surges. Dawn rises of 20-40 mg/dL are normal in non-diabetic individuals and substantially larger in T1D. Exercise exerts a complex bidirectional effect: aerobic activity increases insulin sensitivity for 24-48 hours post-exercise but can cause acute hypoglycemia during the session; anaerobic effort transiently spikes glucose through counter-regulatory hormone release. Any clinically useful forecaster must implicitly encode these time-of-day and

activity-state effects, which motivates the six-hour history window (long enough to span a typical post-prandial response) and the explicit sin/cos hour encoding in the feature set.

2.2 Why Point Predictions Are Dangerous

A scalar forecast can create a false sense of precision. A model that outputs "140 mg/dL in one hour" leaves unspoken whether that estimate is tightly concentrated around the truth or whether it is consistent with a much broader range of plausible outcomes. For an application that compares meals before they are eaten, that distinction matters. A user may make a different choice if told that the central estimate is 140 mg/dL but the upper tail still crosses a clinically undesirable threshold. Forecasting without an interval is therefore not merely incomplete reporting. In this context, it can distort the decision problem itself.

This distinction maps onto the classical split between epistemic and aleatoric uncertainty. Epistemic uncertainty reflects what the model does not know because of limited data or imperfect parameter estimates. Aleatoric uncertainty reflects irreducible variation in the data-generating process, including sensor noise and unmeasured confounders such as stress, sleep, and hormonal state. MC-Dropout is better suited to the first form than the second. Quantile Regression directly models the spread of the conditional target distribution and is therefore especially relevant for noisy physiological time series. The empirical comparison between these two strategies is one of the thesis's central methodological questions (Gal and Ghahramani; Lakshminarayanan et al.).

2.3 Population Heterogeneity and the Case for Personalization

The case for personalization follows directly from the data. Zeevi et al. demonstrated that glycemic response can differ substantially across individuals for the same meal, and even within the same individual across days. Such variation cannot be reduced to calories alone. It emerges from differences in physiology, behavior, timing, and context. A useful forecasting system must therefore be individualized in two senses: it must condition on person-specific recent history, and it must avoid presenting population averages as though they were person-level truths (Zeevi et al.).

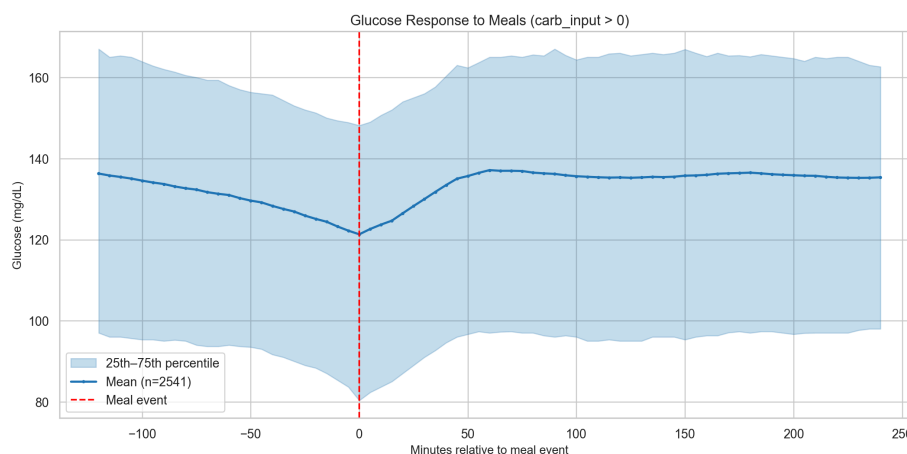


Figure 1. HUPA-UCM means glucose trajectory aligned to reported meal events. The post-meal peak occurs 45-75 minutes after the carb event, biologically grounding the 60-minute forecast horizon.

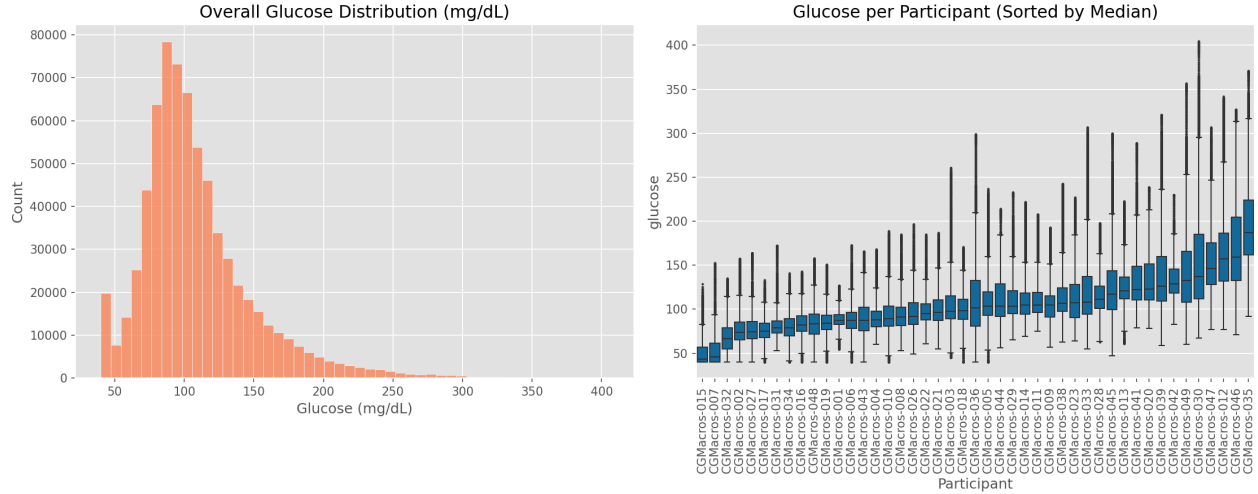


Figure 2. CGMacros per-participant glucose distributions. Healthy, pre-diabetic, and T2D phenotypes are visibly distinct, motivating the need for a model that personalizes rather than averages.

3. Literature Survey and Gap Analysis

3.1 State of the Art in Deep CGM Forecasting

Deep learning for CGM forecasting has evolved from simple feed-forward regressors toward architectures that preserve temporal structure. Recurrent models, especially GRUs and LSTMs, became dominant because they accommodate history explicitly while remaining trainable on relatively small participant cohorts. Martinsson et al. showed that recurrent neural networks can improve glucose forecasting while also supporting variance estimation. More recently, Transformer-based models have entered the literature because self-attention can weigh long-range dependencies without recurrent state. Whether that flexibility helps in CGM forecasting depends strongly on dataset scale and feature richness (Martinsson et al.; Cho et al.; Vaswani et al.).

3.2 Uncertainty Quantification in Deep Learning

Uncertainty quantification remains a weaker point in much of the forecasting literature. Deep Ensembles are often the strongest approach for calibration, but they are computationally expensive and awkward to deploy in a lightweight consumer prototype. MC-Dropout offers a cheap approximation to posterior predictive uncertainty by keeping dropout active at inference time, but its intervals often reflect parameter uncertainty more than the full noise structure of the task. Quantile Regression addresses the problem from a different direction by estimating the conditional distribution directly. In practice, this often yields wider but more honest intervals. The present thesis compares these alternatives on matched forecasting tasks rather than treating uncertainty as a purely theoretical add-on (Lakshminarayanan et al.; Gal and Ghahramani).

3.3 Personalized Nutrition Literature

The personalized nutrition literature provides the conceptual foundation for the project. Zeevi et al. established that individualized glycemic prediction is feasible and clinically meaningful. Subsequent work has expanded the range of features considered, including microbiome, sleep, and behavioral factors. What

remains less developed is the translation from predictive modeling to an everyday decision interface, particularly one that makes uncertainty explicit rather than invisible. Fuel Up+ is positioned at that intersection, between predictive modeling, uncertainty-aware evaluation, and user-facing meal comparison (Zeevi et al.).

3.4 Gap Analysis

Three gaps motivate the work. First, public CGM studies are often evaluated on isolated datasets, which makes cross-dataset claims hard to compare. Second, uncertainty is frequently reported without a serious calibration analysis, especially in prototype systems. Third, consumer-oriented meal tools are typically based on heuristics or descriptive scoring rather than individualized forecasts. Fuel Up+ addresses these gaps by combining cross-dataset harmonization, explicit interval evaluation, and a prototype deployment layer built around meal comparison.

3.5 Research Questions

Drawing from the gap analysis, this thesis investigates four research questions:

- RQ1. On a harmonized three-dataset CGM corpus, does a GRU or a Transformer produce lower out-of-participant MAE, and does the answer depend on dataset size and feature richness?
- RQ2. Does delta-target training (predicting change in glucose) outperform absolute-target training on biologically-dominated versus insulin-dominated cohorts?
- RQ3. Between MC-Dropout and Pinball-loss Quantile Regression, which strategy achieves empirical coverage closer to the 80% nominal target, and at what interval-width cost?
- RQ4. Can the calibrated forecast be translated into a usable consumer product (meal comparison + Safety Score) that is deployable on commodity web infrastructure?

4. Problem Statement and Objectives

4.1 Problem Statement

Given a six-hour window of glucose, meal, and activity history for an individual, the task is to estimate glucose sixty minutes into the future together with an interval that captures meaningful predictive uncertainty, and then to convert that forecast into a ranking of candidate meals that a user can interpret. The system should generalize to unseen participants, operate on commodity hardware, and remain explicit about the fact that it is an advisory forecasting layer rather than a therapeutic controller.

4.2 Objectives

- O1. Harmonize three publicly available CGM datasets (HUPA-UCM, BrisT1D, CGMacros) into a unified five-minute-grid pipeline with documented leakage controls.
- O2. Evaluate GRU and Transformer Encoder architectures in both absolute and delta target modes across three feature scenarios (carbs-only, full-macros, macros-plus-activity).
- O3. Benchmark the deep models against Persistence, Linear Drift, Ridge Regression, and LightGBM baselines, reporting MAE, RMSE, and range-stratified metrics.
- O4. Implement and compare MC-Dropout and Pinball-loss Quantile Regression heads on identical Transformer backbones, measuring coverage (PICP) and interval width (MPIW).

- O5. Deliver a web-based demo dashboard with a Meal Comparison tool and a 0-100 Safety Score, ready for user studies in future work.

5. Scope, Methodology, and Design

5.1 Scope and Delimitations

The scope of the thesis is deliberately bounded. The work concerns advisory glucose forecasting, not insulin recommendation, diagnosis, or closed-loop control. Evaluation is restricted to three public datasets with different cohort types and annotation regimes. Training was carried out on 10% subsamples during iterative development to remain tractable on Apple Silicon hardware, so the reported results should be interpreted primarily as comparative architectural evidence rather than as final performance ceilings. This limitation is discussed directly in Section 7.

Fuel Up+ is organized as a layered system. The data layer ingests raw files from the three source datasets, standardizes units where needed, applies participant-level splits, and generates masks for missingness. The feature layer constructs six-hour windows and appends engineered summaries such as rolling glucose statistics, carbohydrate and insulin aggregates, activity summaries, and time-of-day encodings. The model layer hosts GRU and Transformer backbones in PyTorch. The uncertainty layer either samples MC-Dropout predictions or reads quantile outputs directly. Finally, the application layer exposes the model through a lightweight Flask API connected to a browser-based dashboard.

The three source datasets were chosen for complementarity rather than uniformity. HUPA-UCM provides type 1 diabetes data with glucose, insulin, and wearable signals. BrisT1D offers a pre-windowed competition dataset in mmol/L with strong forecasting utility but less flexibility for feature engineering. CGMacros contributes the richest nutritional annotation, including carbohydrate, protein, fat, fiber, calories, and meal photographs, and covers healthy, pre-diabetic, and type 2 diabetic participants. Together, the three datasets allow the thesis to test whether modeling choices behave differently in insulin-dominated and diet-dominated settings.

The prediction target is glucose at $t+60$ minutes. Each deep model is trained in two modes. Absolute-target training predicts the future value directly. Delta-target training predicts the change from current glucose to future glucose. This distinction matters because delta mode forces the model to learn dynamics rather than relying primarily on the current value. Feature scenarios are then layered on top of that target definition:

Scenario 1 uses glucose and carbohydrates, with insulin in T1D datasets;

Scenario 2 adds full macronutrients for CGMacros;

Scenario 3 adds activity channels on top of the richest available meal representation.

HUPA-UCM (Hidalgo et al.) contains 25 T1D participants from the Hospital Universitario Príncipe de Asturias and Universidad Complutense de Madrid wearing FreeStyle Libre 2 CGMs and Fitbit Ionic smartwatches for ≥ 14 days. BrisT1D (University of Bristol, Kaggle) is a pre-windowed 9-participant

training set plus a 15-participant test set in native mmol/L. CGMacros (Bent et al., PhysioNet) contains 45 participants across healthy, pre-diabetic, and T2D phenotypes, with the richest public nutritional labeling available: carb, protein, fat, fiber, calories, and photographs per meal. For CGMacros the Abbott Libre CGM is adopted as the primary target (0.03% missingness versus 8.4% for Dexcom).

Protecting against leakage is crucial in CGM forecasting because random row-level splits can allow a model to memorize participant-specific baselines. To avoid this, participant-level train, validation, and test partitions are created deterministically using MD5-hash assignment, and all scaling statistics are fit on training data alone. The evaluation protocol further checks that imputation uses only past values, that time encodings derive from the window endpoint rather than the target timestamp, and that no participant appears across splits. These details are not incidental. They determine whether the reported test error reflects genuine out-of-participant generalization.

Figure 3. Sliding-window sanity check on CGMacros S1: five randomly selected windows showing six hours of glucose history and the corresponding 60-minute target. The continuity of the traces supports the integrity of the resampling and window-construction pipeline.

5.5 Split Policy

Participant-level splits are enforced via MD5-hash-based deterministic assignment (~68% train / 16% validation / 16% test). BrisT1D additionally uses GroupKFold ($k=3$) over its 9 training participants. Scaling statistics are computed on train only. No target-time information leaks into features; the STEP-0 evaluation protocol documents ten leakage checks, each verified in code.

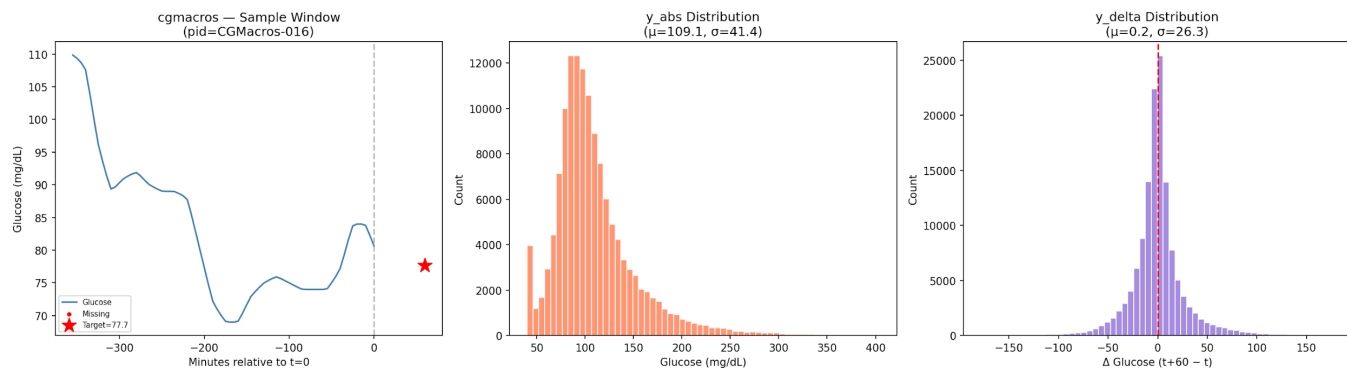


Figure 3. Sliding-window sanity check on CGMacros S1: five random samples overlaying six-hour glucose history (blue) with the 60-minute target (amber). Smooth, biologically plausible trajectories confirm the pipeline’s temporal integrity.

5.6 Uncertainty Heads

Two uncertainty heads were evaluated. In the MC-Dropout setting, dropout remains active during inference and $K = 30$ stochastic forward passes are used to approximate a posterior predictive distribution. In the Quantile setting, the network emits q10, q50, and q90 directly and is trained with the Pinball loss plus a soft monotonicity penalty to discourage quantile crossing. The central comparison is therefore between a lightweight approximate Bayesian strategy and a direct conditional-distribution strategy (Gal and Ghahramani).

6. Work Done: Implementation, Challenges, Mitigations

6.1 Pipeline Engineering

The implementation spans roughly 2,500 lines of Python split across three principal modules: `baselines.py` for data loading and engineered features, `deep_models.py` for GRU and Transformer training, and `uncertainty_models.py` for interval estimation. The system was built in PyTorch, trained on Apple Silicon using Metal Performance Shaders, and wrapped in a small Flask server for interactive demonstration. Although the codebase began as a capstone prototype, by the end of the project it had become a reasonably disciplined research pipeline with reproducible splits and task-specific loaders.

6.2 Key Challenges and Mitigations

Challenge 1: Cross-Dataset Unit and Schema Heterogeneity

The first challenge was heterogeneity of units and schema. HUPA and CGMacros report glucose in mg/dL, whereas BrisT1D is expressed in mmol/L and arrives already windowed into fixed-length columns. The solution was to preserve native units during training, convert only for cross-dataset clinical interpretation, and hide dataset-specific parsing behind a common interface. This reduced avoidable preprocessing error while still permitting a shared evaluation logic.

Challenge 2: CGM Missingness in Free-Living Data

The second challenge was missingness in free-living CGM data. Sensors drop out for ordinary reasons, including bathing, exercise, charging, and wear failure. Treating missing values as zeros is clearly wrong, but unrestricted forward-fill is also misleading because it creates stale observations that look real. The adopted compromise was to forward-fill only short gaps, reject long-gap windows, and provide explicit mask channels so the model can distinguish observed from imputed inputs.

Challenge 3: Participant-Level Leakage Protection

The third challenge was leakage. In time-series health data, inflated performance is easy to obtain if participant identity is allowed to leak across splits or if normalization statistics are estimated globally. The pipeline avoids this by splitting at the participant level, fitting scalers on train only, and inspecting the protocol manually against a checklist. This may sound procedural, but it is precisely the difference between a plausible benchmark and an untrustworthy one.

Challenge 4: Compute Budget on Commodity Hardware

A fourth challenge was to compute the budget. Full sweeps over all datasets and model settings were expensive on available hardware. Subsampling 10% of the training rows was therefore a practical compromise during model development, reducing turnaround time sufficiently to explore architecture and uncertainty settings without losing the comparative structure of the task. Even after subsampling the models took 15+ hours to run. This choice weakens claims about absolute error minima, but it still supports comparative claims about relative model behavior.

Challenge 5: Transformer Overfitting on Small T1D Cohorts

The final major challenge was Transformer overfitting on small T1D datasets. Even with conservative dimensions, dropout, and early stopping, the Transformer tended to fit faster than the available participant diversity justified. Rather than hiding that outcome, the thesis treats it as a

substantive result: in sparse insulin-dominated settings, the GRU's inductive bias toward recency appears better matched to the data regime than self-attention.

6.3 Demo Dashboard

The demonstration layer was built to make the modeling output legible. The dashboard plots the observed six-hour history, the sixty-minute forecast, and the uncertainty band in distinct visual layers. A meal-comparison component then injects candidate meal compositions as counterfactual inputs and returns a ranked Safety Score based on both the predicted central response and the upper-risk bound. The interface is deliberately simple, because its purpose is not visual novelty but interpretability.

7. Results and Discussion

7.1 Headline Results

Table 1 summarizes the principal test-set results. Two broad patterns are clear. First, the GRU in delta mode is the strongest model on the T1D datasets, particularly HUPA S1, where it records 23.22 mg/dL MAE against the Transformer's 27.23 mg/dL. Second, the Transformer becomes much more competitive when the data are richer and the physiology is less dominated by exogenous insulin, as seen in CGMacros S2 and S3. This suggests that model choice in CGM forecasting should be treated as regime-dependent rather than universal.

Dataset	Scen.	Model	MAE	RMSE
HUPA (mg/dL)	S1	GRU-delta	23.22	33.85
HUPA (mg/dL)	S1	Transformer-delta	27.23	37.97
HUPA (mg/dL)	S3	GRU-delta	25.47	35.13
HUPA (mg/dL)	S3	Transformer-delta	26.32	36.11
CGMacros (mg/dL)	S1	GRU-delta	11.32	15.39
CGMacros (mg/dL)	S1	Transformer-delta	10.76	15.28
CGMacros (mg/dL)	S2	GRU-delta	10.92	15.57
CGMacros (mg/dL)	S2	Transformer-delta	10.58	15.07
CGMacros (mg/dL)	S3	GRU-delta	10.20	14.91
CGMacros (mg/dL)	S3	Transformer-delta	10.44	14.84
BrisT1D (mmol/L)	S1	GRU-delta	2.17	2.83
BrisT1D (mmol/L)	S1	Transformer-delta	2.14	2.83
BrisT1D (mmol/L)	S3	GRU-delta	2.09	2.74
BrisT1D (mmol/L)	S3	Transformer-delta	2.11	2.77

Table 1. Deep-model test MAE and RMSE across dataset, scenario, and architecture in delta-target mode.

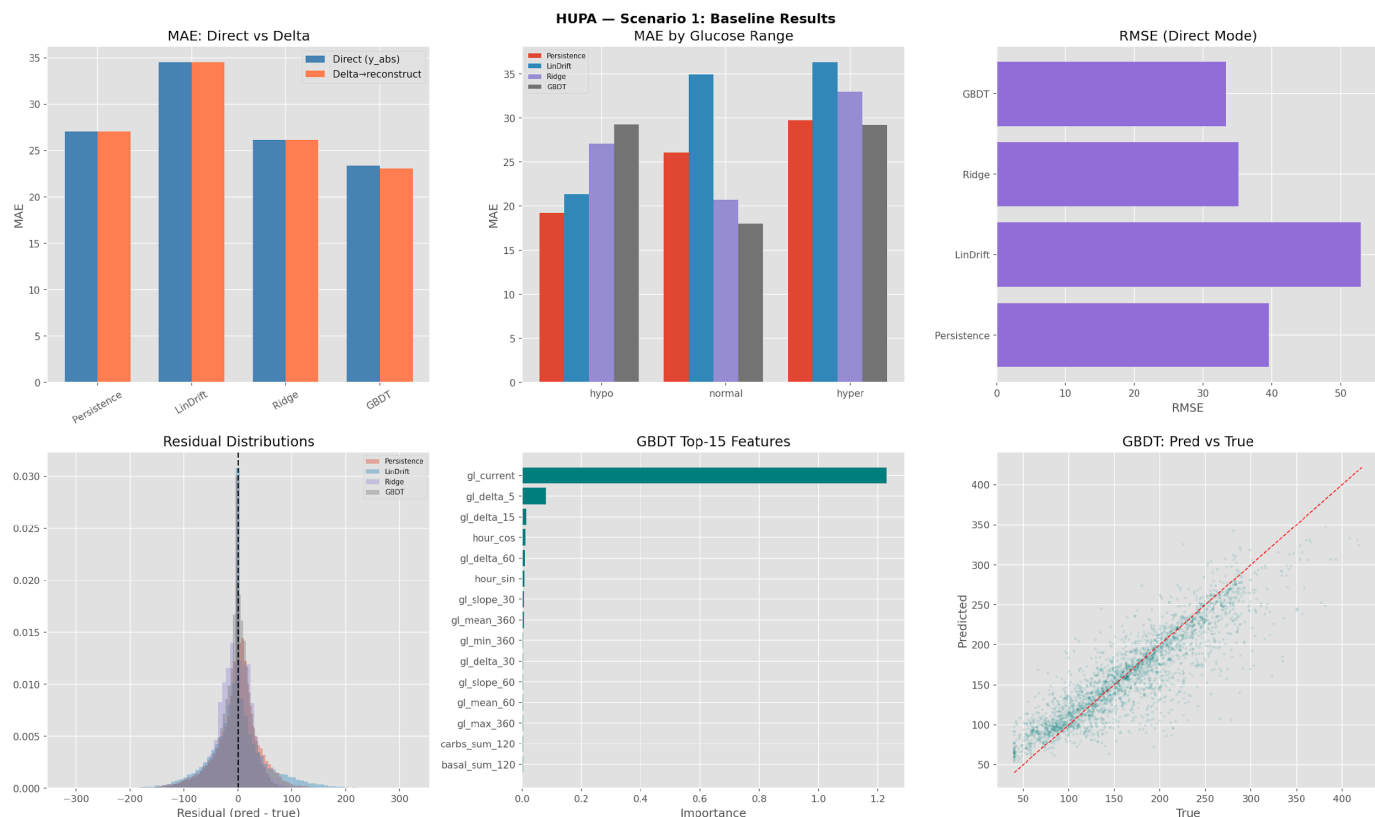


Figure 4. HUPA S1 baseline performance. Persistence MAE 27.06, Ridge 26.16, GBDT 23.10 mg/dL. The GRU-delta (23.22) just matches GBDT, indicating deep models must earn their complexity through uncertainty rather than point accuracy.

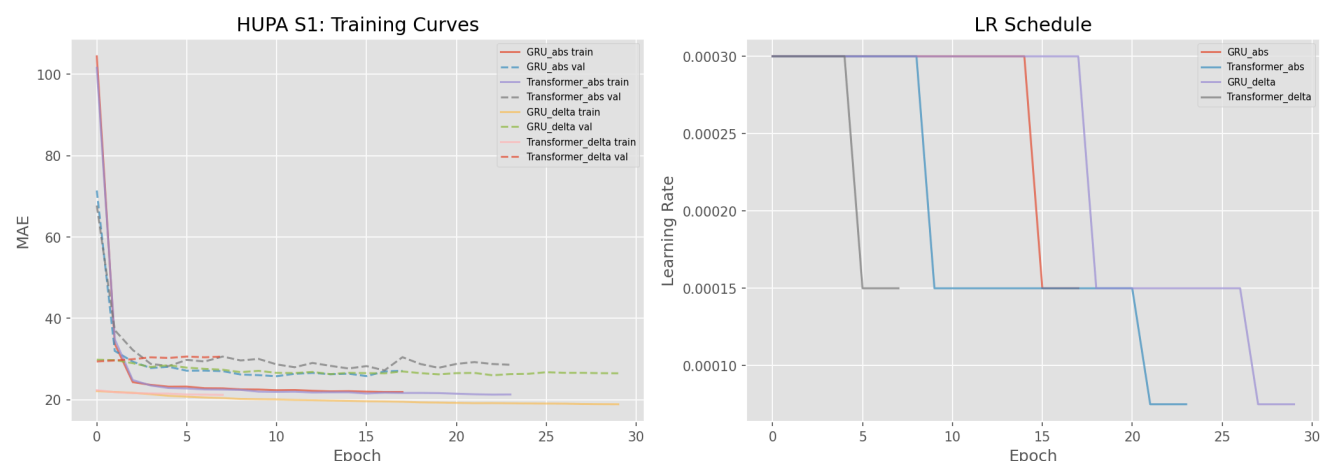


Figure 5. Training and validation loss curves on HUPA S1. The GRU converges smoothly; the Transformer exhibits visible overfitting after epoch 10 despite early stopping, consistent with its parameter count exceeding the effective sample size.

7.2 Answering the Research Questions

RQ1. Which architecture performs better? The answer depends on regime. On the smaller T1D datasets, the GRU is consistently stronger or essentially tied. On CGMacros, the Transformer closes the gap and marginally leads in Scenario 2. The most defensible interpretation is that recurrent gating remains

advantageous when participant count is limited and recent temporal structure dominates, while attention becomes more useful once heterogeneous meal signals are sufficiently rich.

RQ2. Does delta-target training help? In most settings, yes. Delta targets improve performance most clearly on BrisT1D and CGMacros, where predicting change encourages the model to learn dynamics rather than reproduce the current level. On HUPA, the gains are smaller, which is consistent with the greater noise induced by exogenous insulin and sparser participant diversity. Delta mode is therefore not universally transformative, but it is generally the more informative target definition in this task.

RQ3. Which uncertainty method is more reliable? Quantile Regression is the better choice for deployment-oriented forecasting. MC-Dropout produced narrower intervals, but those intervals under-covered badly, including 44.6% observed coverage on HUPA against an 80% nominal target. Quantile intervals were wider, yet materially closer to the desired calibration level. In a meal-support context, that trade-off favors honesty over sharpness.

RQ4. Is deployment plausible? At the prototype level, yes. The dashboard, counterfactual meal comparison, and Safety Score all run on commodity hardware through a simple Flask stack. What is not yet established is clinical utility. The present thesis supports the claim that such a system can be built and benchmarked responsibly. It does not support the stronger claim that the prototype is ready for clinical use.

7.3 Biological Interpretation of the S1 vs. S3 Gap

The gap between Scenario 1 and Scenario 3 is biologically informative. On CGMacros, richer meal and activity signals reduce error because a meaningful share of the glucose trajectory is driven by nutritional composition and everyday behavior. On HUPA, the same additions yield little improvement or even mild degradation, which is consistent with the fact that insulin timing and absorption dominate much of the short-horizon variance in T1D. Additional features do not help when they are weaker than the noise in the primary physiological driver.

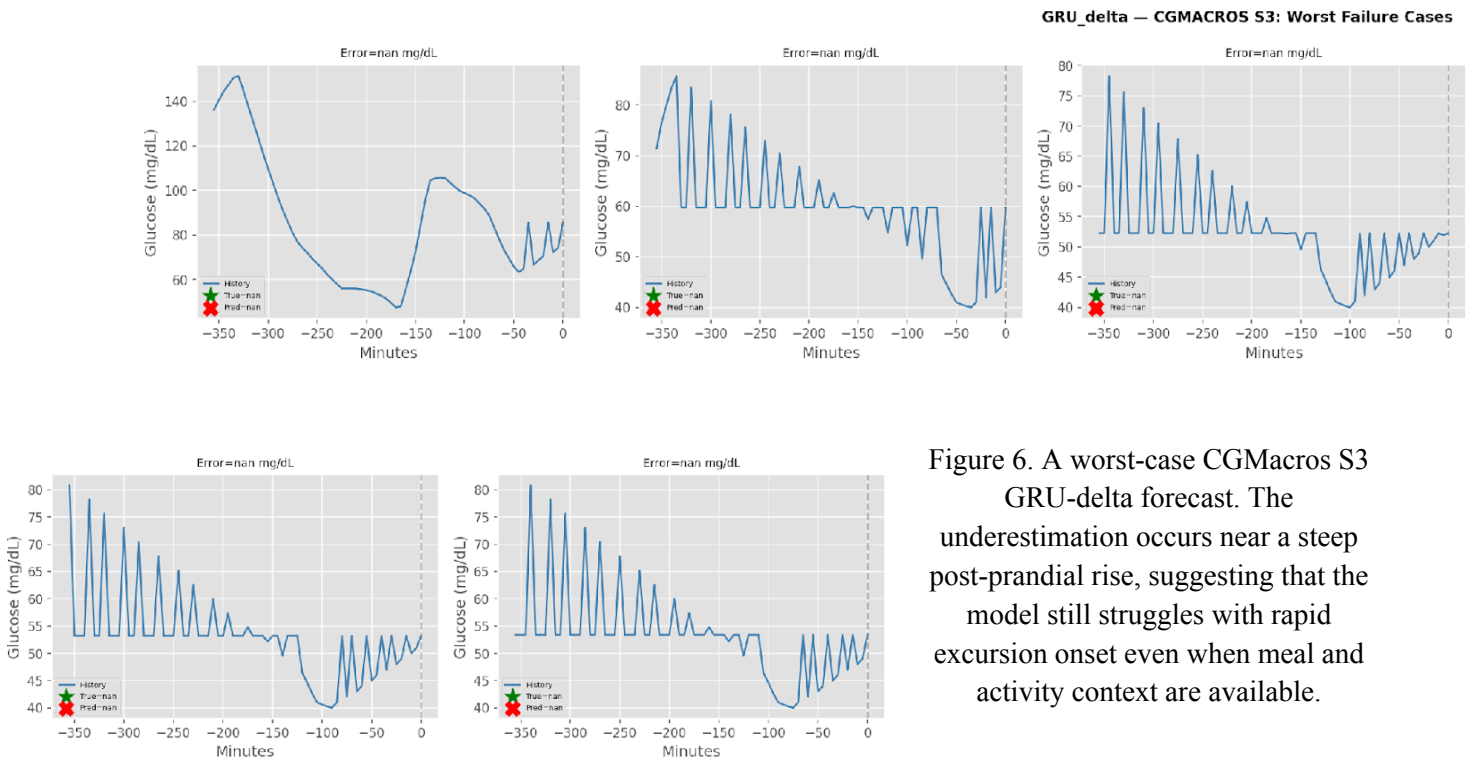


Figure 6. A worst-case CGMacros S3 GRU-delta forecast. The underestimation occurs near a steep post-prandial rise, suggesting that the model still struggles with rapid excursion onset even when meal and activity context are available.

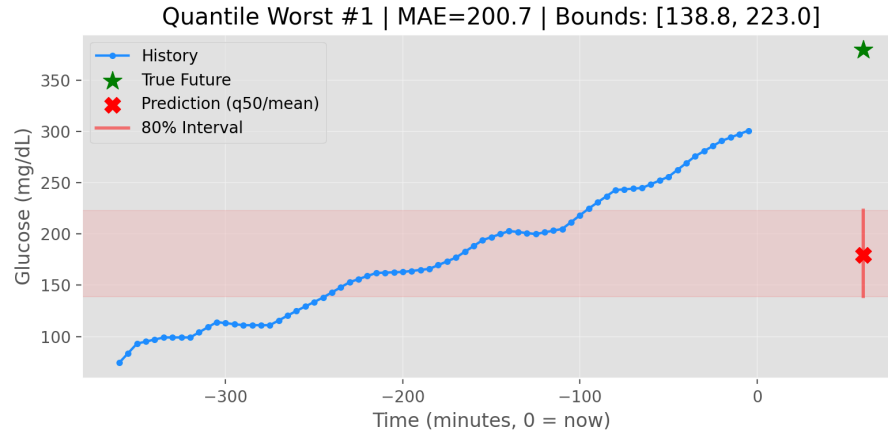


Figure 7. Quantile forecast on a difficult HUPA case. The interval is broad, but it still contains the true future value, illustrating why wider intervals may be more defensible than overconfident narrow ones in high-variance T1D settings.

7.4 Limitations and Threats to Validity

Several limitations should be stated clearly. First, most training was performed on 10% subsamples to make iterative experimentation feasible, so the reported numbers are better interpreted as comparative evidence than as the final attainable accuracy of the architectures. Second, the public datasets differ substantially in cohort composition, annotation detail, and preprocessing assumptions, which complicates one-to-one comparison even after harmonization. Third, the work does not include calibration repair through conformal methods, multi-seed stability analysis, or prospective user evaluation. Finally, none of the datasets is a direct representation of an Indian university cohort, so claims about local deployment remain provisional until local data are collected.

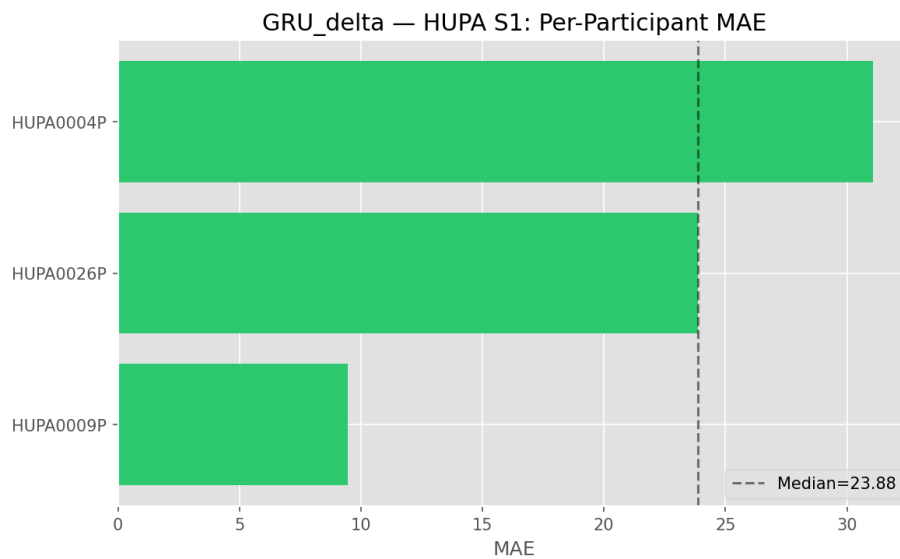


Figure 8. Per-participant MAE for GRU-delta on HUPA S1. The wide spread across held-out participants shows that a single global model remains a compromise and motivates subject-specific fine-tuning as a next step.

8. Conclusions

This thesis makes a bounded but meaningful contribution to personalized metabolic forecasting. It shows that open public CGM datasets can be brought into a common evaluation framework, that architecture choice depends strongly on cohort regime, and that uncertainty quality materially changes the credibility of meal-support predictions. Across the tested settings, the GRU in delta mode emerges as the most practical general-purpose model for T1D forecasting at current data scales, while the Transformer becomes competitive when nutritional context is richer and the data regime is more favorable.

The broader methodological contribution is the insistence that glucose forecasting for end-user decision support should not be evaluated as a point-estimation task alone. In this thesis, the most useful model is not simply the one with the lowest MAE, but the one that combines acceptable error with uncertainty estimates that are honest enough to guide interpretation. By that criterion, Quantile Regression is the stronger deployment candidate. The resulting prototype does not claim clinical readiness. It does, however, demonstrate a credible path from a student nutrition tracker to a more principled forecasting system.

9. Extensions and Future Work

9.1 Near-Term Engineering (0-6 months)

- Full-data retraining on each dataset with 3-5 random seeds to tighten confidence intervals on reported MAE.
- Conformal prediction wrappers on the Quantile head to enforce finite-sample coverage guarantees.
- Model distillation: teacher-Transformer to student-GRU for on-device iOS/Android inference.
- Expanded hyperparameter sweep: attention-head count, hidden dimensions, and longer history horizons (12-hour windows).

9.2 Ashoka University Pilot (6-12 months)

Ashoka University remains a sensible pilot environment because the original product idea emerged from student meal behavior, irregular schedules, and dining-hall decision making. A pilot, however, should begin modestly. The first step is not outcome optimization but calibration and usability: collecting local CGM traces, evaluating whether the model's intervals remain credible on this cohort, and testing whether the Safety Score changes meal selection in a way users actually understand.

9.3 Scaling Across India (12-36 months)

Scaling beyond a pilot requires more than engineering. It requires population-relevant data, careful clinical framing, and regulatory discipline. Indian dietary patterns, body-composition profiles, and metabolic risk distributions differ materially from the public datasets used here, so any wider deployment would require local retraining or at least fine-tuning on an Indian cohort. Just as importantly, claims should grow only after evidence grows, moving from retrospective forecasting to prospective behavior studies and then, only if justified, to longer-term outcome evaluation.

9.4 Hybrid Causal-Neural Architecture

A further research direction is hybrid modeling. A purely data-driven sequence model is flexible, but it has limited explicit knowledge of carbohydrate absorption, insulin-on-board, or physiological

constraints. One promising extension would be to combine the present neural models with a lightweight pharmacokinetic or mechanistic layer so that known structure is retained while flexible residual dynamics are still learned from data. Such a hybrid approach may be especially useful in T1D cohorts where mechanistic timing matters greatly.

References

1. Bent, Brinnae, et al. "CGMacros: A Pilot Scientific Dataset for Personalized Nutrition and Diet Monitoring." *Scientific Data*, vol. 12, 2025, article 1672.
2. Cho, Kyunghyun, et al. "Learning Phrase Representations Using RNN Encoder-Decoder for Statistical Machine Translation." *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Association for Computational Linguistics, 2014, pp. 1724-34.
3. Contreras, Iván, and Josep Vehi. "Artificial Intelligence for Diabetes Management and Decision Support: Literature Review." *Journal of Medical Internet Research*, vol. 20, no. 5, 2018, article e10775.
4. Gal, Yarin, and Zoubin Ghahramani. "Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning." *Proceedings of the 33rd International Conference on Machine Learning*, vol. 48, PMLR, 2016, pp. 1050-59.
5. Hidalgo, J. Ignacio, et al. "HUPA-UCM Diabetes Dataset." *Data in Brief*, vol. 55, 2024, article 110559.
6. Hovorka, Roman, et al. "Nonlinear Model Predictive Control of Glucose Concentration in Subjects with Type 1 Diabetes." *Physiological Measurement*, vol. 25, no. 4, 2004, pp. 905-20.
7. Howey, David C., et al. "[Lys(B28), Pro(B29)]-Human Insulin: A Rapidly Absorbed Analogue of Human Insulin." *Diabetes*, vol. 43, no. 3, 1994, pp. 396-402.
8. Lakshminarayanan, Balaji, Alexander Pritzel, and Charles Blundell. "Simple and Scalable Predictive Uncertainty Estimation Using Deep Ensembles." *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 6402-13.
9. Ma, Jing, et al. "Effects of Fat on Gastric Emptying of and the Glycemic, Insulin, and Incretin Responses to a Carbohydrate Meal in Type 2 Diabetes." *The Journal of Clinical Endocrinology and Metabolism*, vol. 91, no. 4, 2006, pp. 1396-1403.
10. Martinsson, John, et al. "Blood Glucose Prediction with Variance Estimation Using Recurrent Neural Networks." *Journal of Healthcare Informatics Research*, vol. 4, 2020, pp. 1-18.
11. University of Bristol. "BrisT1D Blood Glucose Prediction Competition." *Kaggle*, 2024, kaggle.com/competitions/brist1d.
12. Vaswani, Ashish, et al. "Attention Is All You Need." *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 5998-6008.
13. Zeevi, David, et al. "Personalized Nutrition by Prediction of Glycemic Responses." *Cell*, vol. 163, no. 5, 2015, pp. 1079-94.

Publications and Communications

No peer-reviewed publication has resulted from this work at the time of submission.